

PRACTICAL NO: 9TH

AIM: Hypothesis Testing:

a. Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests, and interpret the results.

b. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

Part A: Hypothesis Testing on a Business Problem

```
print("T110 SANJANA SAHU")
import numpy as np
import pandas as pd
from scipy import stats

np.random.seed(42)

# Step 1: Sample Data (simulate productivity of 40 employees after training)
sample_data = np.random.normal(loc=53, scale=8, size=40) # observed after new program
population_mean = 50 # old known average

# Step 2: Conduct One-Sample t-test
t_stat, p_value = stats.ttest_1samp(sample_data, population_mean)

# Step 2: Conduct One-Sample t-test
t_stat, p_value = stats.ttest_1samp(sample_data, population_mean)

alpha = 0.05

print("=== One-Sample t-Test: Productivity ===")
print(f"Sample Mean      : {np.mean(sample_data):.2f}")
print(f"Population Mean   : {population_mean}")
print(f"t-statistic       : {t_stat:.4f}")
print(f"p-value           : {p_value:.4f}")

# Step 3: Interpretation
if p_value < alpha:
    print("\nResult: Reject the Null Hypothesis (H0).")
    print("Conclusion: The new training program has a statistically significant")
```

OUTPUT:

```
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=== One-Sample t-Test: Productivity ===
Sample Mean      : 51.25
Population Mean   : 50
t-statistic       : 1.0379
p-value           : 0.3057

Result: Fail to Reject the Null Hypothesis (H0).
Conclusion: No significant evidence that the training program
changes productivity.
```

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```
branch_A = np.random.normal(loc=200, scale=15, size=35) # daily sales
branch_B = np.random.normal(loc=210, scale=18, size=35)

t_stat2, p_value2 = stats.ttest_ind(branch_A, branch_B, equal_var=False)

print("=== Two-Sample t-Test: Branch A vs Branch B Sales ===")
print(f"Mean Branch A: {np.mean(branch_A):.2f}, Mean Branch B: {np.mean(branch_B):.2f}")
print(f"t-statistic: {t_stat2:.4f}, p-value: {p_value2:.4f}")

if p_value2 < alpha:
    print("Reject H0 -> Significant difference in sales between branches.")
else:
    print("Fail to Reject H0 -> No significant difference in sales.")
```

```
... === Two-Sample t-Test: Branch A vs Branch B Sales ===
Mean Branch A: 200.05, Mean Branch B: 208.84
t-statistic: -2.5858, p-value: 0.0119
Reject H0 -> Significant difference in sales between branches.
```

Part B: A/B Testing on a Marketing Campaign

```
[5] !pip install statsmodels -q
[4] print("T110 SANJANA SAHU")
import numpy as np
import pandas as pd
from statsmodels.stats.proportion import proportions_ztest
import scipy.stats as stats

# Step 1: Sample Data
# Group A (old page): 1000 visitors, 120 conversions
# Group B (new page): 1000 visitors, 150 conversions
visitors_A, conversions_A = 1000, 120
visitors_B, conversions_B = 1000, 150

# Step 3: Interpretation & Decision
if p_value < alpha:
    print("\nResult: Reject H0.")
    if conv_rate_B > conv_rate_A:
        print("Conclusion: Page B (new design) has a significantly HIGHER")
        print("conversion rate. Recommend deploying Page B.")
    else:
        print("Conclusion: Page A performs significantly better. Retain Page A.")
else:
    print("\nResult: Fail to Reject H0.")
    print("Conclusion: No statistically significant difference between the")
```

OUTPUT:

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=== A/B Test: Landing Page Conversion ===
Page A Conversion Rate: 12.00%
Page B Conversion Rate: 15.00%

Z-statistic: -1.9630
p-value : 0.0496

Result: Reject H0.
Conclusion: Page B (new design) has a significantly HIGHER
conversion rate. Recommend deploying Page B.

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contingency_table = pd.DataFrame({
 "Converted": [conversions_A, conversions_B],
 "Not Converted": [visitors_A - conversions_A, visitors_B - conversions_B]
}, index=["Page A", "Page B"])

chi2, p_chi, dof, expected = stats.chi2_contingency(contingency_table)

print("=== Chi-Square Test (Cross-check) ===")
print(contingency_table)
print(f"\nChi2 Statistic: {chi2:.4f}, p-value: {p_chi:.4f}")

=== Chi-Square Test (Cross-check) ===
 Converted Not Converted
Page A 120 880
Page B 150 850

Chi2 Statistic: 3.6009, p-value: 0.0577

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